# PIZZA

## SALES ANALYSIS USING SQL

# ABOUT ME

Hi, I'm Ayush, an aspiring Data Analyst / Data Scientist passionate about turning raw data into meaningful business insights.
I enjoy working with SQL, Excel, Python, Statistics, and Data Visualization to solve real-world business problems.

# BASIC QUESTIONS

# TOTAL NUMBER OF ORDERS PLACED

```sql
-- Retrieve the total number of orders placed.

SELECT
    COUNT(order_id) total_orders
FROM
    orders;

```

# TOTAL REVENUE GENERATED

```sql
-- Calculate the total revenue generated from pizza sales.

SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_revenue
FROM
    order_details
        JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id;
```

# HIGHEST-PRICED PIZZA

```sql
-- Identify the highest-priced pizza.

SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

# MOST COMMON PIZZA SIZE

```sql
-- Identify the most common pizza size ordered.

SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

# TOP 5 MOST ORDERED PIZZA TYPES

```sql
-- List the top 5 most ordered pizza types along with their quantities.

SELECT
    pizza_types.name,
    SUM(order_details.quantity) AS total_quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY total_quantity DESC
LIMIT 5;
```

# INTERMEDIATE QUESTIONS

# TOTAL QUANTITY OF EACH PIZZA CATEGORY

```sql
-- Join the necessary tables to find the total quantity of each pizza category ordered.

SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS total_quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY total_quantity DESC;
```

# DISTRIBUTION OF ORDERS BY HOUR

```sql
-- Determine the distribution of orders by hour of the day.

SELECT
    HOUR(orders.order_time) Hours,
    COUNT(orders.order_id) order_quantity
FROM
    orders
GROUP BY Hours;
```

# CATEGORY-WISE DISTRIBUTION

```sql
-- Join relevant tables to find the category-wise distribution of pizzas.

SELECT
    pizza_types.category AS Category, COUNT(pizza_types.name)
FROM
    pizza_types
GROUP BY Category;
```

# AVERAGE NUMBER OF PIZZAS PER DAY

```sql
-- Group the orders by date and calculate the average number of pizzas ordered per day.

SELECT
    ROUND(AVG(total_quantity), 0) AS average_total_quantity
FROM
    (SELECT
        orders.order_date AS orders_by_date,
            SUM(order_details.quantity) AS total_quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders_by_date) AS order_quantity_by_date;
```

# TOP 3 MOST ORDERED PIZZA BASED ON REVENUE

```sql
-- Determine the top 3 most ordered pizza types based on revenue.

SELECT
    pizza_types.name AS pizza_names,
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) Revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_names
ORDER BY Revenue DESC
LIMIT 3;
```

# ADVANCED QUESTIONS

# PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE

```sql
-- Calculate the percentage contribution of each pizza type to total revenue.

SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(order_details.quantity * pizzas.price),
                                2) AS total_sales
                FROM
                    order_details
                        JOIN
                    pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
            2) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

# CUMULATIVE REVENUE

```sql
-- Analyze the cumulative revenue generated over time.

WITH sales AS (
    SELECT
        orders.order_date,
        SUM(order_details.quantity * pizzas.price) AS revenue
    FROM order_details
    JOIN pizzas
        ON order_details.pizza_id = pizzas.pizza_id
    JOIN orders
        ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date
)
SELECT
    order_date,
    ROUND(
        SUM(revenue) OVER (ORDER BY order_date),
        2
    ) AS cumulative_revenue
FROM sales;
```

# TOP 3 PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY

```sql
-- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

WITH pizza_revenue AS (
    SELECT
        pizza_types.category,
        pizza_types.name,
        SUM(order_details.quantity * pizzas.price) AS revenue
    FROM pizza_types
    JOIN pizzas
        ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN order_details
        ON order_details.pizza_id = pizzas.pizza_id
    GROUP BY
        pizza_types.category,
        pizza_types.name
)
SELECT
    name,
    revenue
FROM (
    SELECT
        category,
        name,
        revenue,
        RANK() OVER (
            PARTITION BY category
            ORDER BY revenue DESC
        ) AS rn
    FROM pizza_revenue
) ranked
WHERE rn <= 3;
```

# THANK YOU